

Assignment 3

1.  **Worth:** 6%
2.  **Due Date:** See LEA
3.  **Late submissions:** 10% penalty per late day.
4.  **Submission:**
 - Put all `.py` file(s) into a `.zip` and submit on LEA
 - Please use the provided files **WITHOUT** renaming them.
 - Fill the docstring header at the top of each `.py` file with your name, section and date.
 - Use type hinting on all functions.
 - Use the `my_functions.py` to define all your functions
 - Use the `assignment3.py` file to test all your functions.

Learning Objectives

- Break down a problem into smaller solvable steps
- Design and implement an algorithm using the concepts covered in class.
- Use `for` loops for repetition
- Use `if`, `if - else` and `elif` statements for conditional logic
- Implement and use **functions** to improve code structure and reuse logic.

Question 1 : Taxes

Write a function to help a person estimate their net income.

◆ **Function:** `net_income()`

- **Input parameter:** `hourly_rate` and `hours_per_week`
- **Returns:** the `net_income` which is calculated as `gross_income * (1-tax_rate)`

Important Notes

- Convert the **weekly** salary to a **yearly** salary.
- Assume that there is **52 paid** weeks in a year.
- `hourly_rate` and the `hours_per_week` can have decimal values.
- Use `if/elif/else` statements to determine the appropriate tax rate.
- Use the [Quebec income tax rates](#) below to find the `tax_rate` associated with the income range:

Income range	Tax rate
\$51,780 or less	14%
More than \$51,780 but not more than \$103,545	19%
More than \$103,545 but not more than \$126,000	24%
More than \$126,000	25.75%

Testing

In your main script, test your functions with different `hourly_pay` and `weekly_hours` .

For example, if

```
hourly_pay = 14.50
weekly_hours = 15
```

The expected output should be:

```
Your estimated net income is: 9726.60 $
```

Question 2: Active Substance

Context

The half-life of a substance is the time it takes for its concentration to be reduced to half.

In medicine, this determines how often a patient should take a drug. Physicians decide on a dosage schedule by multiplying the half-life by 4 or by 5 which gives a **rough** estimation of the next time the patient should take the medication.

The concentration of a drug in the bloodstream is described by the following formula:

$$C(t) = C_0 e^{-\lambda t}$$

Here λ (lambda) is the decay constant specific to each drug, is calculated using this formula:

$$\lambda = \frac{\ln(2)}{t_{1/2}}$$

For example the drug Donepezil used to treat dementia, has a half-life of 70 hours (4,200 minutes). According to the 4-5 * half-life rule used by doctors, this means the drug should be taken every 11-14 days.

The objective is to use `python` to write a script which will determine the intake frequency of a medication with at an accuracy +/- 1 hour.

◆ **Function: `concentration_percent()`**

- **Input parameters:** the `time` in **hours** and the `half_life` of the substance in **hours**
- **returns** the `concentration` in percentage using this formula:

$$\frac{C(t)}{C_0} = e^{-\lambda t}$$

where λ is given by this formula:

$$\lambda = \frac{\ln(2)}{t_{1/2}}$$

Important:

- Note: the time and the half-time can be fractional values
- Use `math.log()` for the $\ln()$ and `math.e` for the Euler's constant e

◆ **Function: `intake_frequency()`**

Determines how often a drug should be taken based on when its concentration falls below 5%.

- **Input parameter(s):** the `half_life` of a substance (in hours)
- **returns:** the time interval between each dose.

Algorithm

- set the `lower_bound` $\leftarrow \text{floor}(4 * t_{1/2})$
 - set the `upper_bound` $\leftarrow \text{ceil}(5t_{1/2})$
 - Initialize the `intake_hour` $\leftarrow \text{upper_bound}$
 - If the `upper_bound` ≤ 1 , return the `intake_hour`
 - For every hour t between `lower_bound` to `upper_bound`
 - Compute $\frac{C(t)}{C_0}$
 - If $\frac{C(t)}{C_0} \leq 5\%$
 - set the `intake_hour` $\leftarrow t$
 - break
 - return the `intake_hour`
- **Important :**
 - The `range()` only takes input values of type `int` . Round the start and end values of the interval using the appropriate rounding functions.
 - Each iteration, use `concentration_percent()` to get the remaining concentration in percentage relative to the initial concentration C_0 .

Testing

Compare your function's output with the analytical solution for various half times of various substances.

$$t_{end} = \frac{\ln(20)}{\ln(2)} * t_{1/2}$$

Substance	Half life
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Substance	Half life
Caffeine	~5 hours
Theobromine (found in chocolate)	~7 hours
Ibuprofen (Advil)	~1.8 hours
Acetaminophen (Tylenol)	2 hours

÷ Question 3: Greatest Common Divisor

In mathematics, the **greatest common divisor (GCD)**, also known as **greatest common factor (GCF)**, of two or more integers, which are not all zero, is the largest positive integer that divides each of the integers.

For example, the GCD of 8 and 12 is 4, that is, $\text{gcd}(8, 12) = 4$

[wikipedia](#)

◆ Function: gcd()

- **Input parameters :**

- a (int)
- b (int)

- **Returns:** the greatest common divisor

Algorithm

Write additional code that calculates the greatest common divisor using the following algorithm:

Repeat **enough** times:

- if $a \leq b$, then $b \leftarrow b - a$
- if $b < a$, then $a \leftarrow a - b$
- if a or b is equal to zero, **print** a and b and **break** out of the loop

The gcd will be either a or b , whichever is not zero

Testing

In the main script, call the function with various values, for example:

```
print(gcd(105,252))
```

Expected Output:

```
21, 0
```



Question 4: Hexagon t-shirts

A company makes fabrics based on the trendy hexagon pattern shown below:



They are currently offering a wide selection of sizes characterized by the number of blue, white and green hexagons (`num_hex`) as shown in the figures below.



`num_hex = 6`

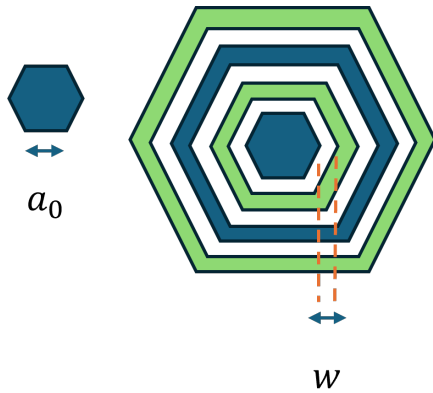


`num_hex = 7`



`num_hex = 8`

All patterns have a central hexagon of side length a_0 . The concentric hexagons are evenly spaced by a value of w .



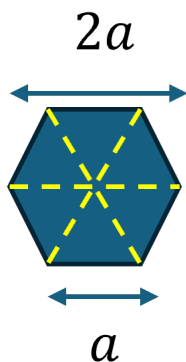
The company has tasked you with creating a script to calculate the area covered by the green fabric given the number of hexagons, the central hexagon's side length and the spacing between consecutive hexagons.

◆ Function: `hex_area()`

- **Input parameter :**
 - `side` : Length of a side of the hexagon
- **Returns:**
 - Hexagon's area using the formula:

$$\frac{3\sqrt{3}}{2}a^2$$

Assume that all hexagons are regular and where a is the side of the hexagon as shown in the figure below:



◆ Function: `green_area()`

- **Input parameter:**
 - `num_hex` : total number of hexagons of the printed pattern

- a_0 : the length of side of the central hexagon
- w : the spacing between consecutive hexagons w

- **Returns:** Area covered by the green fabric.

Hint: Start by determining the relationship between two consecutive hexagons' side length.

Testing

In the main script, test your function with various values:

```
print("Area for 6 hex: ", green_area(6,3.0,1.0))
print("Area for 7 hex: ", green_area(7,3.0,1.0))
print("Area for 8 hex: ", green_area(8,3.0,1.0))
```

Grading Ruberic

Criteria	Description	Points
Name and Id	Docstring file descriptions are completed on each file	1
Code Quality	Use of variables and constants to avoid magical numbers. Naming conventions are respected. Import statements are defined at the top of each file. Code is easy to read and equations are broken down when necessary. All functions are defined at the same indentation level. The <code>main()</code> is defined and used in the.	5
Conditional statements	Correct and optimal use of comparison operators and conditional statements. Avoids redundant conditions.	10
Functions	All functions are defined in the <code>my_functions.py</code> script. Correct calculations are performed Type hinting is correctly used in each function	10

Criteria	Description	Points
Loops	Correct use of loops and break statement when necessary	15
Logic & algorithm	Correct translation of a problem into logical and broken down steps to achieve the goal	20
Comments	Uses comments to explain logic when necessary.	5

- Note the ponderation of each criteria might vary from one question to question.